

Professional Issues

1 Professional Issues Report

1.1 Introduction

Professional issues arise where computing systems handle real user data, particularly when that data is sensitive. In this project, the most relevant issue is the privacy and security of financial information within a web-based asset management platform. The system stores user identity alongside data relating to digital asset holdings and overall portfolio value, which, if exposed, could leave users vulnerable to targeted scams or exploitation.

Although authentication was initially implemented as a functional requirement, its importance became clearer as the system developed. Implementing token-based security and protecting backend endpoints highlighted that privacy and security are not optional features, but core responsibilities in systems where users must trust that their financial data is handled safely.

This also introduces an inherent trade-off, as increasing security often adds complexity to both system design and user interaction, requiring careful balance between usability and protection.

1.2 Nature of the Professional Issue

Data privacy and security are critical in systems handling financial information, as such data reveals far more than basic personal details. While general user data identifies an individual, financial data exposes what they own, making it more valuable and potentially harmful if misused.

The most relevant risks in this context are targeted scams and loss of user trust. If users cannot rely on a system to protect their data, its core purpose becomes irrelevant. A breach would not only have immediate consequences, but could also lead to long-term reputational damage and reduced confidence in the platform.

This highlights a key challenge in financial systems, where the value of the data increases the incentive for malicious exploitation, making security not just a

technical concern, but a risk management problem.

This project reinforced that financial data cannot be treated as ordinary information. Its misuse has real-world implications, making privacy and security fundamental considerations in any such application.

1.3 Relevance to the Project

This issue is directly relevant due to the type of data handled. The system stores user identity information alongside financial data, including asset holdings, transaction history, portfolio value, and user-defined goals. Together, this forms a clear representation of a user's financial position.

All data is stored within a PostgreSQL database and accessed through the backend, with authentication enforced on every request. External API calls are also handled server-side, ensuring that API keys are not exposed to the frontend and that data can be validated before use.

Despite these protections, risks remain. The system depends on correct access control to prevent users from accessing data that is not their own, and reliance on third-party services such as Firebase introduces an external dependency. This reinforces that privacy and security are not abstract concerns, but central to the system's design and operation.

While these measures strengthen security, they also introduce reliance on correct implementation, meaning that any flaws in access control logic could undermine the entire system despite the presence of authentication mechanisms.

1.4 Addressing Privacy and Security in the System

Security was implemented through a combination of authentication, backend control, and data validation. Firebase authentication ensures that only authenticated users can access protected endpoints, with all requests validated before processing. On the frontend, an authentication context is used to manage user state and enforce protected routes, preventing unauthenticated users from accessing restricted views.

Although this approach improves security and control, it also increases backend complexity, requiring careful design to ensure that performance and scalability are not negatively affected.

A backend-first approach was adopted, ensuring that all sensitive logic and data handling occur on the server. This prevents exposure of API keys and allows the system to control how data is retrieved and returned. The frontend is limited to presentation, reducing the risk of exposing sensitive operations.

User ownership validation ensures that individuals can only access their own data, while caching mechanisms reduce excessive API calls and improve system

stability. Together, these decisions create a controlled environment in which security is applied consistently across the application.

1.5 Limitations and Risks

Despite these measures, several limitations remain. The system relies on third-party services such as Firebase, introducing a dependency that is outside direct control. Any issues with these services would require reactive handling.

Security testing was limited to functional validation, meaning the system has not been evaluated against more advanced attack scenarios. As a result, vulnerabilities such as token misuse or misconfigured endpoints may still exist.

Additionally, design decisions such as caching, while beneficial for performance, introduce the potential risk of stale or inconsistent financial data if not carefully managed.

Additionally, the system lacks advanced protections such as encryption at rest and monitoring mechanisms, which are expected in production-level financial applications. For this reason, the system would not yet be suitable for real-world deployment, highlighting the gap between an academic implementation and a fully production-ready system.

1.6 Ethical and Legal Considerations

Developers have a responsibility to protect user data, particularly in systems involving financial information. Releasing a system with known security weaknesses would be unethical, as it exposes users to potential harm.

In this project, user data is not shared with third parties and is used solely to provide system functionality. This aligns with principles of minimising data use and respecting user privacy.

Legally, financial systems must comply with regulations such as GDPR, which require the secure handling of personal data. Failure to meet these standards can result in fines, legal consequences, and loss of user trust.

Furthermore, relying on third-party services such as Firebase raises questions around shared responsibility, as security is partially delegated to external providers rather than being fully controlled within the system itself.

1.7 Impact on Development Process

Privacy and security significantly influenced the development process, particularly in shaping a backend-first architecture. This ensured that all critical logic and data handling remained under controlled conditions.

Authentication and API handling introduced additional complexity, requiring each request to be validated and linked to a specific user. This made development more challenging, particularly in the early stages, but reinforced the importance of integrating security from the outset.

The project also highlighted that backend design is critical, as many features depend on secure and well-structured data handling. If repeated, more time would be spent understanding security practices in advance to improve efficiency and reduce trial-and-error.

1.8 Conclusion

Data privacy and security are essential in systems that handle financial information. This project demonstrated that protecting user data is not simply a feature, but a responsibility that directly shapes system design.

The development process reinforced the importance of backend control, secure data handling, and user trust. It also changed how financial applications are viewed, highlighting that beyond the interface, significant effort is required to ensure systems are reliable and secure.

Overall, this project demonstrates that professional issues such as privacy and security must be considered throughout development in order to produce systems that users can confidently rely on.

In financial systems, failures in privacy and security are not merely technical issues, but failures of trust with real-world consequences. This highlights that effective system design must extend beyond functionality, requiring careful consideration of risk, responsibility, and the impact of design decisions on end users.